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# On the Polar-Wander Path for Australia during the Cenozoic

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*Summary*

Detailed palaeomagnetic measurements potassium-argon and (K–Ar) dating have been carried out on continuous lava sequences in the Nandewar, Liverpool and Barrington Volcanoes of New South Wales. Consistent K–Ar ages in each case indicate that the volcanoes were built rapidly at $17{\cdot}5 \pm 0{\cdot}3$ m.yr (Nandewar), $33{\cdot}7 \pm 0{\cdot}7$ m.yr (Liverpool) and $51{\cdot}6 \pm 0{\cdot}7$ m.yr (Barrington), where the errors quoted are the standard deviations. Most of the 150 flows investigated show stable directions of magnetization after magnetic cleaning. Several field reversals are noted and in one instance a series of nine flows records the transition zone from reversed to normal polarity. The frequency and timing of the reversals are consistent with the Cenozoic geomagnetic time scale deduced from the interpretation of marine magnetic anomalies in terms of the sea-floor spreading hypothesis around 17 and 34 m.yr, but not at 52 m.yr. It appears that the extrapolations made in its derivation may no longer be valid at that time. The three new precisely determined palaeomagnetic poles add detail to the Cenozoic polar-wander path for Australia. A steady movement of the South Pole at about 0·45° per m.yr ($\sim 5$ cm yr$^{-1}$) is indicated for the past 100 m.yr. This movement is not consistent with a simple northward movement of Australia as inferred from sea-floor spreading data. It indicates in addition a westward excursion of the South Pole, which is attributed to a component of polar wandering superimposed upon the continental drift.

## 1. Introduction

Previous palaeomagnetic studies have shown that the palaeolatitude of south-east Australia was high (70°–80° for Canberra) from the Late Carboniferous to the Middle Cretaceous (Irving, Robertson & Stott 1963; Irving 1966). The present latitude of Canberra is 35° S, indicating that there has been substantial polar wandering and/or continental drift since the Middle Cretaceous. The idea that much of the latitude change of Australia since the Cretaceous is because of continental drift has been reinforced by the work of Le Pichon & Heirtzler (1968). From the observed magnetic anomalies across the mid-oceanic ridge between Australia and Antarctica, and their interpretation on the basis of the sea-floor spreading hypothesis (Hess 1962; Vince & Matthews 1963), these authors have inferred that Australia moved northwards relative to Antarctica during the Cenozoic.

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The present study was undertaken to examine these possibilities by determining the manner in which the palaeolatitude of Australia changes during the Cenozoic. To compare the land-based polar changes with the ocean-floor drift south of Australia requires that a series of poles from well-dated rocks be determined with considerable accuracy (polar errors of 5° or less). Previous palaeomagnetic measurements are generally inadequate for this purpose, either because the age of the rocks was not known precisely or because large polar errors resulted from sampling at only a few localities.

Twenty sites in the Older Volcanics of Victoria have been studies by Mumme (1963) following a preliminary investigation by Green & Irving (1958). On palaeontological grounds these basalts are thought to be Palaeoceilto Eocene in age (Cookson & Dettman 1959; Singleton & Joyce 1969). Irving, Stott & Ward (1961) investigated eight sites from the Tertiary basalts of New South Wales. The localities sampled are now known to range in age from about 15 to 45 m.yr (McDougall & Wilkinson 1967; Stipp & McDougall 1968; and other unpublished information). Volcanics in southern Queensland have been studied at 11 sites by Robertson (1966). Webb, Stevens & McDougall (1967) report the age of these rocks as ranging from 25 to 21 m.yr. The Newer Volcanics of Victoria have been investigated by Green & Irving (1958), Parry (1960) and McDougall, Allsopp & Chamalaun (1966). These latter authors found their K–Ar age ranged from 5 to 0·5 m.yr, and other evidence indicates activity until about 5000 years ago.

A considerable number of Cenozoic shield volcanoes occur near the eastern margin of Australia and these provide suitable sequences of lava flows both for detailed palaeomagnetic work and K–Ar dating. In this paper combined palaeomagnetic and dating measurements are reported from three large shield volcanoes in northern New South Wales. These are the Nandewar, Liverpool and Barrington Volcanoes (Fig. 1). Their numerous thin flows are relatively well exposed, allowing systematic collections of samples to be made. By working on thick sequences of lava flows the internal consistency of K–Ar dates is checked, and a reliable age can be found. The palaeomagnetic data also provides information on the frequency of reversals of the geomagnetic field at various times during the Cenozoic. This information can then be compared with the geomagnetic polarity time scale deduced for the Cenozoic from sea-floor spreading measurements (Heirtzler *et al.* 1968).

## 2. K–Ar ages and geology

### 2.1 *Sampling*

In each volcano the lava sequences were sampled from top to base for palaeomagnetism. Only samples from the fresher flows were used for dating. In the Liverpool Volcano a single section gave reasonable coverage whereas in the other two volcanoes incomplete sections had to be collected. For the one road section (Mount Kaputar road, Nandewar Volcano), a portable water-cooled diamond drill was used to obtain 2·2 cm diameter, 12 cm long cores. In the other sections one or two hand samples were collected from the lower more massive parts of each flow. Flows were collected irrespective of any apparent alteration.

### 2.2 *Methods of K–Ar dating*

Details of the K–Ar dating techniques used have been given by McDougall (1964, 1966) and by Cooper (1963). A rock sample is fused in a high vacuum system and after a tracer of argon is added, all gases except argon and other rare gases are removed using hot copper oxide, hot titanium and a molecular sieve. The isotopic composition of the argon is determined using a Reynolds-type mass spectrometer,

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and the amount of radiogenic argon calculated from this. Potassium is measured on a flame photometer using a lithium internal standard.

Total rock samples were used in all the age determinations. Prior to dating the samples were examined in thin section, and those with excessive alteration of the potassium-rich phases were rejected. The apparent suitability of the remaining samples was assessed as follows (Tables 1–3). The best rocks (A) showed no alteration of any potassium bearing phases, other (B) showed partial devitrification of glass or alteration of less than 5 per cent of the potassium bearing phases, while a few (C) showed alteration along cracks, as well as alteration of minerals and devitrification of glass.

The reproducibility of the ages depends on both the reproducibility of the potassium and that of the radiogenic argon. Using the duplicate results given in this paper, the coefficient of variation (c.v.) for a single measurement of potassium is found to be 0·3 per cent, and for argon 1·1 per cent. The reproducibility of the radiogenic argon determinations is entirely due to the uncertainty in the tracer calibrations (c.v. 1·5 per cent). From these figures the coefficient of variation expected for ages calculated from a single argon determination and two potassium determinations is 1·15 per cent, and that for duplicate argon and potassium determinations is 0·83 per cent. The errors on the apparent ages given in Tables 1–3 are for the 0·68 probability level (one standard deviation) and they are determined from the uncertainties in the physical measurements of the individual run. The decay constant used in the calculations are $\lambda_\beta = 4{\cdot}72 \times 10^{-10}\ \mathrm{yr}^{-1}$, $\lambda_e = 0{\cdot}584 \times 10^{-10}\ \mathrm{yr}^{-1}$, $^{40}\mathrm{K} = 1{\cdot}19 \times 10^{-2}$ at.per cent. The overall accuracy of dates depends on the uncertainty in the decay constants, the relative abundance of the isotopes of potassium, and also on the absolute calibration of the $^{38}\mathrm{Ar}$ tracer. The total effect of these factors is thought to be less than 5 per cent.

### 2.3 *Nandewar Volcano*

Nandewar Volcano lies 410 km north-west of Sydney on the edge of the plains of the Darling River (Fig. 1). Although now extensively eroded, the volcano was originally elongate with a length of 60 km, a width of 30 km and a total thickness near 800 m. Jenson (1907) described the petrology and structure of the complex lower part of the volcanic pile and suggested that basalt eruptions probably occurred throughout much of the Miocene and Pliocene. Abbot (1969) described the volcano as of shield type, built by trachyandesites and less voluminous olivine basalts, basalts and trachytes, intruded by plugs, sills and dykes of trachyte and alkali rhyolite. The shield was built by a ' succession of quietly extruded flows with no large erosional breaks '. Stipp & McDougall (1968) dated several flows and intrusions using the K–Ar method, and they found ages ranging from 20·5 to 17 m.yr (Middle Miocene, Geological Society of London 1964).

The best section in the Nandewar Volcano is in a steep gully at the head of Lindesay Creek. It is directly below the electricity sub-station between Mount Kaputar and Governor Plug. The section consists of three thick flows (mean thickness 30 m) overlying 64 thin flows (mean thickness 4 m) with gravels of uncertain age at their base. The upper flows are mostly trachyandesites with one trachyte and a few basalts, whilst the lower flows are hawaiite. Most of the flows consist of a massive lower half and a clinkery upper half.

The Mount Kaputar road section is poorly exposed and cannot be precisely correlated with the other section. The sequence is apparently similar except that at the base occurs a sandstone of Triassic age (Abbot 1969). The three thick upper flows are exposed as continuous prominent bluffs down most of the southern flank of the volcano. They appear to have an initial dip of about 5° and to be unfaulted and unfolded. In contrast at the base of the volcanic pile there are extensive intrusions,

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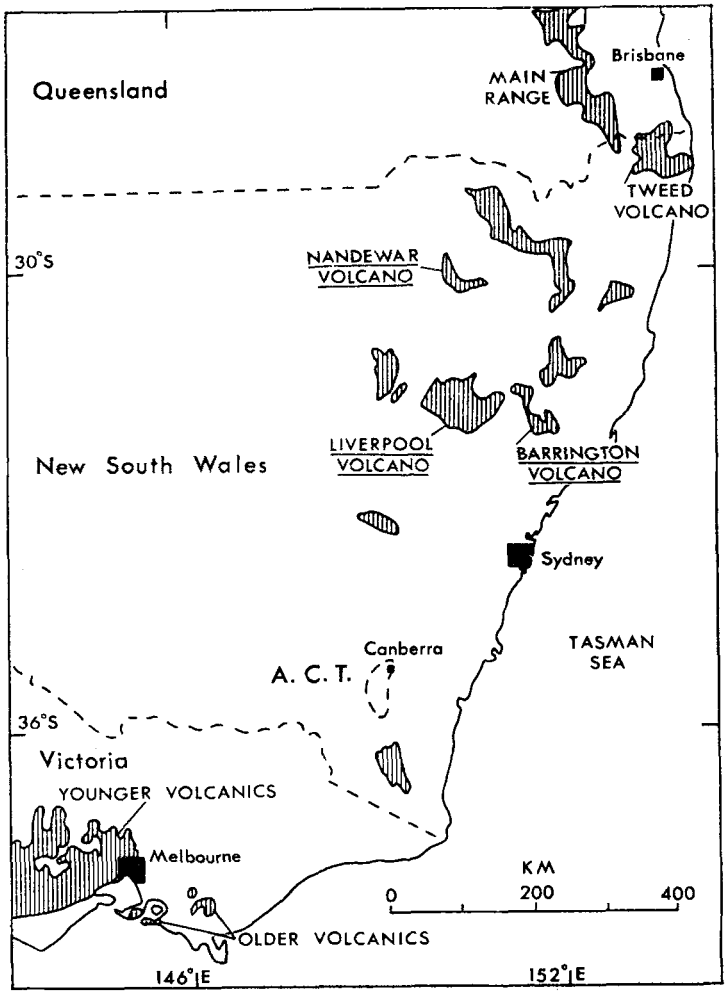



FIG. 1. Map of eastern Australia showing the locations of Cenozoic volcanoes. Those studied in this paper have their names underlined.

and the underlying sediment is locally faulted and titled (Jenson 1907). Although the faults may extend for some distance into the volcanic pile, and some intrusions extend to the top, there is no evidence in the Mount Kaputar area of tilting of the lava flows.

Six flows in the southern part of the volcano, from the sequence in the area examined palaeomagnetically, have been dated (Table 1). These results include one determination by Stipp & McDougall (1968) who also dated a single flow in the northern part of the volcano. The ages listed in Table 1 from the southern part range from 17·2 to 18·0 m.yr and have a mean of 17·5 m.yr with standard deviation 0·3 m.yr. Assuming these flows are of the same age, their coefficient of variation (1·4 per cent) is not significantly different ($P = 0{\cdot}05$) from that expected from experimental error (0·8 per cent). There is an indication that the ages might decrease with increasing altitude, but this cannot be established from the data. It seems unlikely that the flows in the section studied palaeomagnetically would have been extruded over a period greater than 0·6 m.yr (two standard deviations).

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**Table 1**

*Nandewar Volcano: Potassium–argon ages on whole rock samples*

| Sample no. | K (wt per cent) | Rad$^{40}$ Ar ($10^{-11}$ moles/g) | Atm$^{40}$ Ar (per cent) | Calculated age (m.yr) | Mean calculated age (m.yr) | Alteration | Altitude (m) |
|---|---|---|---|---|---|---|---|
| GA2943 | 2·59 | 8·04 | 5·9 | 17·40±0·2 | 17·4±0·2 | B | 1363 |
| | 2·58 | 8·03 | 4·2 | 17·38±0·2 | | | |
| GA2940 | 4·24 | 13·08 | 5·8 | 17·28±0·2 | 17·2±0·2 | B | 1347 |
| | 4·24 | 13·00 | 2·7 | 17·20±0·2 | | | |
| GA2941 | 2·95 | 9·20 | 4·4 | 17·40±0·2 | 17·4±0·2 | B | 1317 |
| | 2·97 | | | | | | |
| GA2942 | 2·91 | 9·20 | 3·3 | 17·68±0·2 | 17·6±0·2 | B | 1251 |
| | 2·91 | 9·15 | 4·4 | 17·58±0·2 | | | |
| GA1534* | 2·19 | 6·83 | 45·3 | 18·08±0·5 | 18·0±0·4 | | 1070 |
| | 2·11 | 6·74 | 58·2 | 17·86±0·5 | | | |
| GA1914 | 2·08 | 6·49 | 39·3 | 17·44±0·3 | 17·5±0·3 | B | 925 |
| | 2·08 | 6·57 | 38·5 | 17·66±0·3 | | | |

* From Stipp & McDougall (1968).

A whole rock trachyte date is reported by Stipp & McDougall (1968)(GA1536, 17·4±0·4 m.yr) from a dome that intrudes sediments just below the lower flows of the Mount Kaputar section. The date is consistent with the intrusion being the same age as part of the flow sequence. Stipp & McDougall (1968) also report an age of 20·5±0·5 m.yr for a single flow from the northern part of the volcano (GA1530). This is significantly older than the ages from the southern part. This could be either because the northern part of the volcano is significantly older, or because of the presence of excess radiogenic argon (Livingston *et al.* 1967) in the anorthoclase phenocrysts used for dating.

### 2.4 *Liverpool Volcano*

The Liverpool Range is an erosional remanant of a shield volcano (here called the Liverpool Volcano) built of alkali olivine basalt lavas. The volcano lies 220 km north of Sydney (Fig. 1) and is part of the main divide of eastern Australia. Lavas total 850 m in thickness (Galloway 1967), and extends east–west for 100 km and north-south for 80 km, while plugs are common only in the southern portion (Dulhunty 1937). By comparing the petrology and geomorphology of the basalts with those of other areas, Sussmilch (1923) and Browne (1933) concluded that the age of the basalts was probably late Tertiary, while David (1950) using similar reasoning thought that the age was early Tertiary.

The only section studied is up a tributary of Yarraman Creek (below Bundulla Trigonometrical Station) and is in the central western part of the volcano. Alkali olivine basalt in flows of 1–8 m thick (mean 6 m) forms the whole 750 m section. No plugs or sills were seen in the area. The basalts range from coarse to fine-grained; they are variable in colour and percentage of amygdales, and show no apparent trend of amount of alteration of flow type with height in the section. The more massive flows, which are exposed for a considerable distance around the hillsides, are continuous and flat lying, so that there has almost certainly been no deformation. This conclusion is supported by the unfaulted, shallow basin shape of the base of the volcanics (Galloway 1967).

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**Table 2**

*Liverpool Volcano: Potassium–argon ages on whole rock samples*

| Sample no. | K (wt per cent) | Rad$^{40}$ Ar ($10^{-11}$ moles/g) | Atm$^{40}$ Ar (per cent) | Calculated age (m.yr.) | Mean calculated age (m.yr.) | Alteration | Altitude (m) |
|---|---|---|---|---|---|---|---|
| GA2915 | 1·43 | 8·60 | 41·1 | 33·65±0·6 | 33·5±0·7 | B | 1160 |
| | 1·42 | 8·54 | 58·7 | 33·42±0·7 | | | |
| GA3466 | 1·18 | 6·53 | 24·3 | 31·01±0·4 | 31·0±0·4 | B | 955 |
| | 1·17 | | | | | | |
| GA3467 | 1·16 | 7·24 | 10·4 | 34·81±0·4 | 34·6±0·4 | B | 851 |
| | 1·16 | 7·18 | 10·0 | 34·45±0·4 | | | |
| GA2913 | 0·97 | 5·84 | 18·4 | 33·56±0·6 | 33·5±0·5 | B | 665 |
| | 0·98 | 5·82 | 28·1 | 33·48±0·4 | | | |
| GA2945 | 1·41 | 8·47 | 9·8 | 33·43±0·4 | 33·3±0·4 | B | 547 |
| | 1·42 | 8·41 | 7·1 | 33·18±0·4 | | | |
| GA2944 | 0·85 | 5·37 | 16·4 | 35·38±0·5 | 34·3±0·5 | B | 468 |
| | 0·84 | 5·07 | 49·6 | 33·41±0·5 | | | |
| | | 5·18 | 30·7 | 34·12±0·4 | | | |
| GA2916 | 0·97 | 5·76 | 14·1 | 33·24±0·6 | 32·9±0·5 | B | 400 |
| | 0·96 | 5·64 | 17·0 | 32·57±0·4 | | | |

The section up a tributary of Yarraman Creek has been dated in detail (Table 2), the ages having a mean of 33·3 m.yr with standard deviation 1·2 m.yr. The coefficient of variation (3·5 per cent) is much larger than can be attributed to errors in measurement, and is due mainly to the young apparent age for GA3466. This sample may have lost radiogenic argon, and if it is rejected then the mean age becomes 33·7 m.yr with standard deviation 0·7 m.yr. The coefficient of variation (2·0 per cent) is still significantly larger ($P = 0{\cdot}05$) than that expected from experimental errors (0·83 per cent). As there is no apparent trend in age with stratigraphic position, there appears to have been some gain or loss of potassium or argon, from these samples. However, the general concordancy of the results suggests that the lavas were extruded over a relatively short period of time. The maximum time covered by the section is unlikely to exceed twice the standard deviation, i.e. 1·4 m.yr.

### 2.5 *Barrington Volcano*

North of the Hunter Valley, 220 km from Sydney is the Mount Royal Range, a deeply disected basalt volcano, with a flat top about 1700 m above sea level. The remnants of this volcano, here called the Barrington Volcano, extend 80 km north–south, 30 km east–west, and are about 700 m thick. The relatively flat top is not caused by flows abnormally resistant to erosion (as in the Liverpool Volcano) but is best explained as an older land surface, possibly a partly stripped upper surface of the volcano. The petrology of the rocks has been described by Benson (1911) and by Browne (1928). Using geological considerations Sussmilch (1923) inferred a Late Miocene or Early Pliocene age for the volcano, while later David (1950) inferred an early Tertiary age. Using the K–Ar method of dating, McDougall & Wilkinson (1967) found an Early Eocene age for a single flow (48·7±0·5 m.yr).

Faulting and possible tilting of the basalts has been discussed by Sussmilch (1940) and more recently by Galloway (1967). From contours at the base of the basalt, Galloway inferred a fault along the line of the lower part of Stewarts Brook, and warping and faulting along the watershed at the northern, western and southern edges of the Barrington Tops plateau. Granite is exposed 500 m above the base of the basalt in the Barrington Tops area so there is either strong warping and/or

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faulting, or a former hill of granite has been covered by the basalt. As all the flows were continuous and flat-lying, it seems unlikely that there has been either significant warping or faulting in the area studied palaeomagnetically. The interpretation of the granite inlier as an exhumed hill is therefore preferred.

Two sections, 21 km apart, were studied in the southern part of the Barrington Volcano. The Stewarts Brook section is a composite formed from seven flows cropping out in Stewarts Brook, where it begins just north of Barrington Trignometrical Station, and 16 flows cropping out in and around a stream flowing north from Meehans Peak into Stewarts Brook. The 500 m of basalt sampled is underlain by 200 m of poorly exposed basalt, that in turn rests on folded Palaeozoic sediments. Basalts with large pyroxene phenocrysts form the middle part of the section. The Sempill Creek section is in a stream just north of Sempels Rock homestead. The well-exposed 300 m section sampled is overlain by 140 m of poorly exposed basalt. From their heights relative to the basement, it is probable that the Sempill Creek section is stratigraphically below, or slightly overlapping the lower part of the Stewarts Brook section. This is consistent with the ages and palaeomagnetism discussed below.

The basalts are variable in grain size, percentage of amygdales and in the proportion of large olivine or pyroxene phenocrysts. The flows usually consist of a grey, columnar jointed lower part, grading into about 1 m of clinkery material at the top. They range in thickness from 8 to 36 m, with a mean near 15 m.

Five flows have been dated through each of the sections at Sempill Creek and Stewarts Brook (Table 3). The result from a flow 4 km from Sempill Creek (McDougall

**Table 3**

*Barrington Volcano: Potassium–argon ages on whole rock samples*

| | Sample no. | K (wt per cent) | Rad$^{40}$ Ar ($10^{-11}$ moles/g) | Atm$^{40}$ Ar per cent | Calculated age (m.y.r.) | Mean calculated age (m.y.r.) | Alteration | Altitude (m) |
|---|---|---|---|---|---|---|---|---|
| Stewarts Brook | GA2924 | 1·05<br>1·05 | 9·74 | 4·1 | 51·53±0·7 | 51·5±0·7 | A | 1550 |
| | GA2925 | 1·05<br>1·05 | 9·57 | 4·9 | 50·69±0·6 | 50·7±0·6 | A | 1550 |
| | GA2949 | 0·99<br>0·99 | 7·95<br>7·91 | 13·4<br>19·4 | 44·76±0·6<br>44·53±0·6 | 44·6±0·6 | B | 1280 |
| | GA2929 | 1·20<br>1·20 | 10·94 | 21·7 | 50·81±0·7 | 50·8±0·7 | A | 1116 |
| | GA2948 | 1·44<br>1·47 | 13·74<br>13·66 | 17·2<br>18·2 | 52·41±0·7<br>52·09±0·7 | 52·2±0·7 | A | 878 |
| Sempill Creek | GA3465 | 1·25<br>1·26 | 10·65 | 46·7 | 47·25±0·6 | 47·3±0·6 | B | 1330 |
| | GA2903 | 1·12<br>1·12 | 10·33 | 5·2 | 51·25±0·9 | 51·3±0·9 | A | 1184 |
| | GA2923 | 0·93<br>0·92 | 7·08<br>7·24 | 16·5<br>11·1 | 42·57±0·7<br>43·48±0·6 | 43·0±0·7 | B | 1124 |
| | GA2902 | 0·93<br>0·94 | 6·93<br>7·66 | 20·3<br>6·1 | 41·20±0·7<br>45·60±0·8 | 43·4±0·8 | B | 1124 |
| | GA2901 | 0·30<br>0·30 | 2·83<br>2·77 | 43·0<br>19·7 | 52·88±0·9<br>51·79±0·7 | 52·3±0·8 | B | 917 |
| | GA2900 | 0·40<br>0·39 | 3·56 | 92·8 | 50·02±4·7 | 50·0±4·7 | C | 885 |
| | GA1961* | 0·86<br>0·87 | 7·59 | 35·9 | 48·7±0·5 | 48·7±0·5 | | 1260 |

*From McDougall & Wilkinson (1967).

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& Wilkinson 1967: GA1961) has also been included in Table 3. Of the 11 flows dated, seven give ages in good agreement from 50 to 52 m.yr, whilst the remaining four range from 43 to 49 m.yr. The younger ages could be due to loss of radiogenic argon from some of these rocks, but it is possible that some of the flows are younger intrusives. Benson (1911) inferred the presence of intrusive sills (theralites and olivine dolerites) below Barrington Trigonometrical Station from the large grain size of some sheet-like bodies. The ' theralites ' below Barrington Trigonometrical Station are near the middle of the section and they have the appearance in the field of a sequence of six thick lava flows. The one sample dated (GA2929) falls in the group of seven older ages. The samples giving the younger ages come from thick planar bodies that appear to be fine-grained, unweathered lava flows. There is no evidence to suggest that the bodies are intrusive.

All samples with degree of alteration category (A) fall into the older group of ages, whereas the younger group of ages are all of category (B). The older group however include some of category (B) and (C) as well. It may be that the younger ages are explained in terms of alteration, but this would be surprising since all samples dated from the Nandewar and Liverpool Volcanoes fall into category (B) and in each case consistent ages were obtained. Loss of radiogenic argon appears to be the most likely explanation for the younger ages from the Barrington Volcano, rather than a period of intrusive activity. If this is assumed to be the case and if the results from GA2900 is excluded because of its large error, then the remaining six ages have a mean of 51·6 m.yr with standard deviation 0·7 m.yr. The coefficient of variation (1·4 per cent) is not significantly different ($P = 0{\cdot}05$) from that expected from experimental errors. This suggests that the flows were erupted over a relatively short period of time, that is unlikely to exceed 1·4 m.yr (two standard deviations).

### 2.6 *Summary of K–Ar ages*

Table 4 summarizes the results of the K–Ar dating of the three volcanoes. The geological periods are those which conform to the Geological Society of London's time scale (Geological Society of London 1964).

## 3. Palaeomagnetism

### 3.1 *Orientation and measurement*

Samples were oriented with sun-compass and level or in some cases with a prismatic compass and level. From each hand sample between three and nine 2·2 × 2·2 cm cylinders were drilled in the laboratory. The standard deviation of the errors in the orientation marks on the finally prepared specimens is estimated to be about 3°.

The direction and intensity of the natural remanent magnetization of each specimen was measured on an astatic magnetometer (Irving 1956). The magnetometer has a standard deviation in measurement of 1° in direction and 2 per cent in intensity, and a lower limit of measurement of $7 \times 10^{-5}$ e.m.u. cm$^{-3}$. A more sensitive magnetometer was used for a sample from one trachyte flow.

**Table 4**

*Summary of K–Ar results*

| Volcano | Mean age and s.d. | Geological period* | Estimated maximum time for eruption = 2 × s.d. |
|---|---|---|---|
| Nandewar | 17·5 ± 0·3 m.yr | Middle Miocene | 0·6 m.yr |
| Liverpool | 33·7 ± 0·7 m.yr | Lower Oligocene | 1·4 m.yr |
| Barrington | 51·6 ± 0·7 m.yr | Lower Eocene | 1·4 m.yr |

* Following the time scale of Geological Society of London (1964).

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### 3.2 *Alternating field demagnetization*

A test specimen from each lava flow was progressively demagnetized in five or more steps up to a maximum of 600 Oe peak value using an alternating field demagnetization apparatus similar to that described by McElhinny (1966). Initial changes in direction usually occurred in fields up to 100 Oe. Thereafter, if for three or more successive demagnetizing fields, changes in direction were less than 5°, the specimen was regarded as having reached a stable end-point. The procedure generally follows the criteria for magnetic cleaning outlined by As & Zijderveld (1958) and McElhinny & Gough (1963). Several specimens from each sample were then demagnetized in a field slightly higher than that at which the stable end-point was first reached. Specimens exhibiting no stable end-point as defined above were classified ' unstable ' and the measurements were rejected. After magnetic cleaning the mean direction of magnetization of all specimens from each sample was calculated using the statistical method of Fisher (1953). Mean flow directions were then calculated from sample directions in the same way.

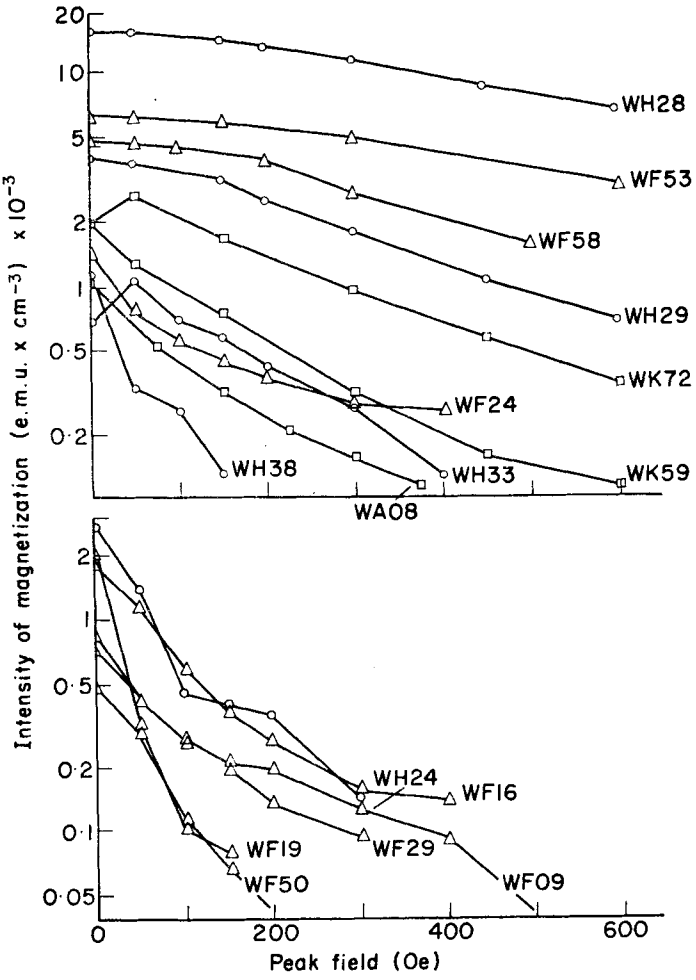



FIG. 2. Typical alternating field demagnetization curves for stable (top) and unstable (bottom) basalts from the three Cenozoic volcanoes. ○, Liverpool Volcano; □, Barrington Volcano; △, Nandewar Volcano.

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The response of samples from all three volcanoes to progressive a.f. demagnetization is very similar. Initial intensities vary widely over several orders of magnitude in each volcano. Stable specimens show approximately exponential decrease of intensity with increasing field (Fig. 2a), the intensity generally dropping by about one-half between each step. This type of response is typical of thermoremanent magnetization observed in many basalts (e.g. Stott 1963; Van Zijl, Graham & Hales 1962). Examples of the response of the ' unstable ' specimens (as defined above) are shown in Fig. 2(b). In general the initial intensities of these specimens is lower than the average and they are characterized by a more rapid decrease in intensity up to 100 Oe applied field. The definition of unstable specimens used in this study is rather more stringent than that normally used, but in spite of this only eight out of 151 flows had to be rejected on these grounds.

### 3.3 *Nandewar Volcano*

Of the 67 flows in the Lindesay Creek section, it was possible to sample only 41. Because it is probable that the older flows of the volcano are not represented in this section, 14 of the lower flows exposed on the Mount Kaputar road were also sampled. The initial intensities of magnetization of the specimens range from $1{\cdot}7\times10^{-5}$ to $6{\cdot}3\times10^{-3}$ e.m.u. $cm^{-3}$. In general, small low coercivity components were removed at 50 Oe, and thereafter the directions of magnetization changed by

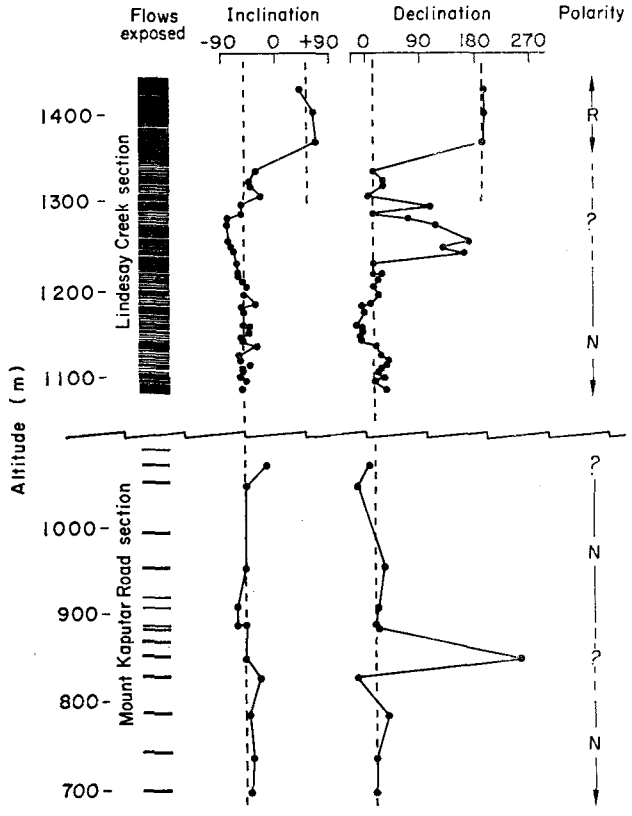



FIG. 3. Nandewar volcano. Variation in declination and inclination through the sections sampled. Dashed lines show the overall mean direction.

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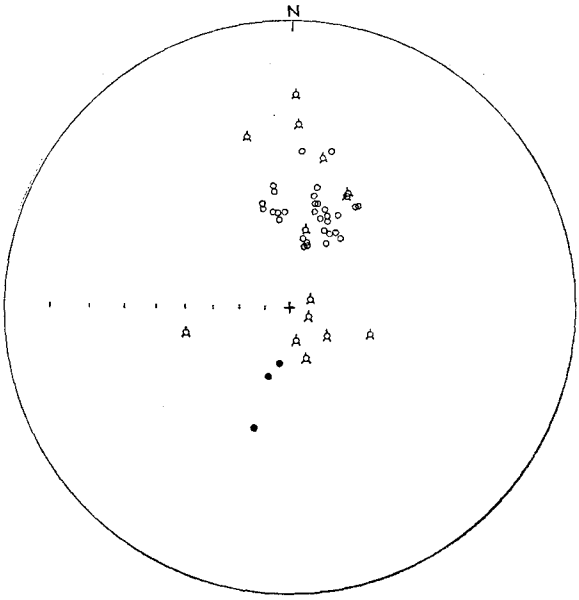



FIG. 4. Nandewar volcano. Stereographic projection of directions of magnetization of all stable flows. ●, Lower hemisphere; ○, upper hemisphere; starred circles are those not used in calculating the overall mean as explained in the text.

less than 4°. The field selected for magnetic cleaning was not critical. Five specimens were classified as being unstable and for two flows the mean directions of two cores collected from each of them differed by more than 40°. Results from these flows were not considered further.

The mean directions for the remaining 48 flows are illustrated in Figs 3 and 4, the former showing changes of declination and inclination in stratigraphical order. The section in Lindesay Creek can be divided into three parts, a lower normally magnetized region, an upper reversely magnetized region, and a central region in which the directions are variable. Over a period covered by the extrusion of seven flows, the declination shows marked deviations from its previously steady value whilst the inclination becomes slightly steeper. Normal polarity is maintained. These anomalous directions are interpreted as due to an increase in the non-dipole filed relative to the main dipole field just prior to a reversal taking place. Disturbances in field directions just prior to a change in polarity have also been observed in previous records (Van Zijl *et al.* 1962; Watkins 1965) and is also expected on theoretical grounds (Allan 1962). Since the purpose of this investigation is to determine the direction of the average main dipole field at various times with the best precision possible, the 11 flows which precede the upper reversely magnetized ones have therefore been rejected in calculating the overall mean. In the Mount Kaputar road section three directions are similarly interpreted, but in this case no reversal of the field is recorded in the section.

Of the 34 flows which remain, three are reversely magnetized and the rest are normal. The means of these two groups differ by $174 \pm 21°$, which is not significantly different from 180°. The 34 flows are believed to cover sufficient time (of the order of $10^5$ yr estimated earlier) to average out secular variation. The mean direction and corresponding pole are given in Tables 5 and 6 respectively, and this represents a well-defined position at 17·5 m.yr.



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**Table 5**

*Summary of palaeomagnetic results*

| | Age (m.yr) | $N$ | $D$ | $I$ | $R$ | $k$ | $\alpha_{95}$ | $\delta_0$ |
|---|---|---|---|---|---|---|---|---|
| Nandewar Volcano (lat. 30·3S, long. 150·2E) | 17·5 | 34 | 192·4 | +53·0 | 33·23 | 43·14 | 3·7 | 12·1 |
| Liverpool Volcano (lat. 31·7S, long. 150·2E) | 33·7 | 36 | 200·4 | +59·2 | 35·23 | 45·49 | 3·5 | 11·8 |
| Barrington Volcano (lat. 32·0S, long. 151·4E) | 51·6 | 33 | 193·0 | +65·5 | 32·33 | 48·47 | 3·6 | 11·4 |

$N$, Number of flows; $D$, declination in degrees east of true north; $I$, inclination in degrees measured positive downwards; $R$, resultant of $N$ unit vectors; $k$, precision parameter (Fisher 1953); $\alpha_{95}$, radius of circle of 95 per cent confidence about mean direction; $\delta_0$, angular standard deviation.

### 3.4 *Liverpool Volcano*

Samples were collected from 49 of the flows exposed in the tributary of Yarraman Creek. Initial intensities of magnetization ranged from $10^{-4}$ to $2 \times 10^{-2}$ e.m.u. cm$^{-3}$. Only one sample was classified as unstable, the remainder being stable according to the criteria previously outlined. Changes in inclination and declination in stratigraphic order are illustrated in Fig. 5. In the lower half of the section the flows are mainly reversely magnetized, whereas in the upper half they are normally magnetized. Three reversals of the geomagnetic field are recorded in the

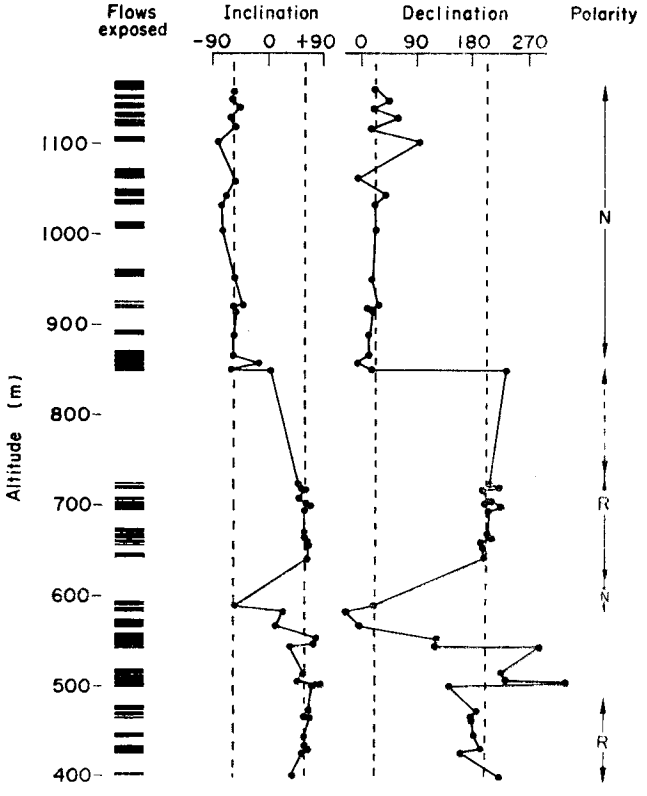



FIG. 5. Liverpool volcano. Variation in declination and inclination through the sections sampled. Dashed lines show the overall mean direction.

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## Table 6

*Summary of Australian late Mesozoic and Cenozoic palaeomagnetic poles*

| | Formation | K-Ar Age (m.yr) | *N* | Long. E | Lat. S | $A_{95}$ (deg.) | λ (Canberra) (deg.) | Reference |
|---|---|---|---|---|---|---|---|---|
| 1 | Victorian Younger Basalts (lat. 38·0° S, long. 143·5° E) | 4·5–0·5 | 11 | 236·8 | 82·5 | 7·7 | 35 | McDougall *et al.* (1966)<br>McDougall *et al.* (1966) |
| 2 | Nandewar Volcano (lat. 30·3° S, long. 152·2° E) | 17·5±0·3 | 34 | 0·84·3 | 78·5 | 4·5 | 39 | This paper<br>This paper |
| 3 | South-east Queensland (lat 27·0° S, long. 152·2° E) | 25–22 | 11 | 086·1 | 78·5 | 11·7 | 39 | Robertson (1966)<br>Webb *et al.* (1967) |
| 4 | Liverpool Volcano (lat. 31·7° S, long. 150·2° E) | 33·7±0·7 | 36 | 095·5 | 71·1 | 4·8 | 45 | This paper<br>This paper |
| 5 | Barrington Volcano (lat. 32·0° S, long. 151·4° E) | 51·6±0·7 | 33 | 125·6 | 70·5 | 5·3 | 52 | This paper<br>This paper |
| 6 | Victorian Older Basalts (lat. 38·0° S, long. 145·5° E) | Palaeocene-Eocene* | 20 | 144·2 | 61·1 | 16·5 | 63 | Mumme (1963)<br>Singleton & Joyce (1969) |
| 7 | Mount Dromedary Intrusion (lat. 36·3° S, long. 150·7° E) | 93 | 22 | 139·2 | 55·5 | 8·2 | 68 | Robertson (1963)<br>Evernden & Richards (1962) |
| 8 | Cygnet Alkaline Intrusion (lat. 42·8° S, long. 147·1° E) | 98±3 | 15 | 158·0 | 53·4 | 11·1 | 73 | Robertson & Hastie (1962)<br>McDougall & Leggo (1965) |
| 9 | Tasmanian Dolerites (lat. 42·0° S, long. 147·0° E) | 170 | 51 | 160 | 51 | 6·0 | 72 | Irving (1963)<br>McDougall (1961) |

*N*, Number of sites; $A_{95}$, radius of circle of 95 per cent confidence about mean pole position; λ, palaeolatitude (95 per cent confidence limits are same values as $A_{95}$).

* From palaeontological considerations.



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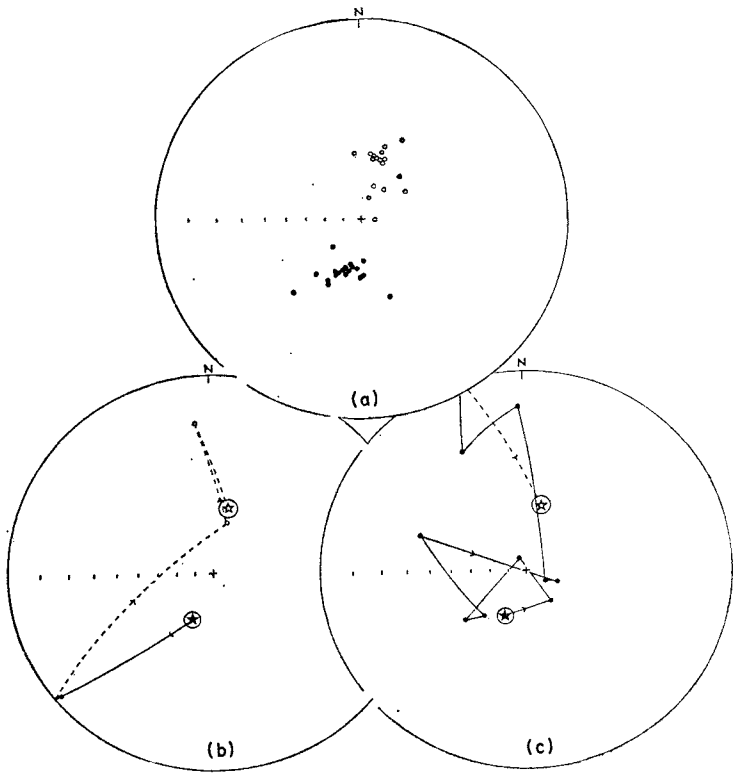



FIG. 6. Liverpool volcano. Stereographic projections of directions of magnetization of all stable flows (circles). (a) All flows excluding those occurring in the transition zones shown in (b) and (c). (b) Transition zone covering three flows between 851 and 860 m altitude. (c) Transition zone covering nine flows between 500 and 585 m altitude. ●, Lower hemisphere; ○, upper hemisphere. The stars are the mean normal and reversed directions of (a) together with their circles of 95 per cent confidence.

section. In two of these a transition zone is observed. Between 500 and 585 m altitude there are nine flows which record the transition from reversed to normal. The directions follow a fairly random path (Fig. 6c) as has been observed in previous records of transition zones (Van Zijl *et al.* 1962; Watkins 1965). In the second case the transition zone occurs over only three flows between 851 to 860 m altitude, the path being shown in Fig. 6(b). The remaining flow directions are shown in Fig. 6(a) from which the mean direction has been calculated.

The means of the normal and reversed groups of Fig. 6(a) differ by $178 \pm 7°$ which does not differ significantly from the expected 180°. It seems likely, therefore, that the section covers more than sufficient time to average out secular variation. The mean direction and pole position in Tables 5 and 6 respectively, therefore represent a well-defined position at 34 m.yr.

### 3.5 *Barrington Volcano*

Samples were collected from 13 flows in the Sempill Creek section and 23 flows in the Stewarts Brook section. Initial intensities range from $3 \times 10^{-4}$ to $2 \times 10^{-2}$ e.m.u. $cm^{-3}$. Under a.f. demagnetization samples showed considerable change in both direction

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and intensity between 0 and 50 Oe. At higher fields the direction was essentially constant and the intensity decreased by one-half every 150 Oe. All flows were stable with the exception of one near the base of Stewarts Brook section.

In Fig. 7 directions of magnetization are plotted as a function of altitude. The Sempill Creek section is predominantly reversely magnetized, but a single normally magnetized flow near the base suggests that the polarity of the field changed twice. The Stewarts Brook section contains two periods of reverse polarity separated by a period of normal polarity. Two directions are anomalous as is obvious from Fig. 8, which shows the flow mean directions of magnetization. The stratigraphic relationship of the two sections is not certain, but it seems probable that the Stewarts Brook section overlies that in Sempill Creek as shown in Fig. 7. An analysis of the mean normal and reversed directions of Fig. 8 (excluding the two anomalous directions) shows that the two groups differ by $170 \pm 6°$, which is just significantly different from 180°. This would appear to indicate that either the normal or reversed groups taken separately do not by themselves cover sufficient time to average out secular variation. However the mean of all the 33 flows, combined irrespective of sign, will represent a time interval roughly twice that of either group. It could well be that sufficient time is now represented by all 33 flows to average out secular variation over both of the sections. The mean direction and pole position calculated in Tables 5 and 6 respectively, therefore represents a further well-defined position at 52 m.yr.

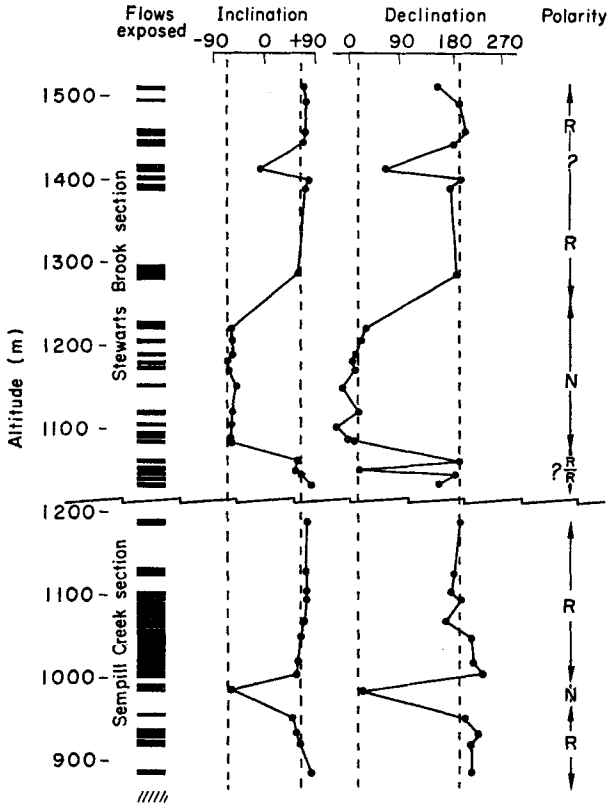



FIG. 7. Barrington volcano. Variation in declination and inclination through the section sampled. Dashed lines show overall mean direction.

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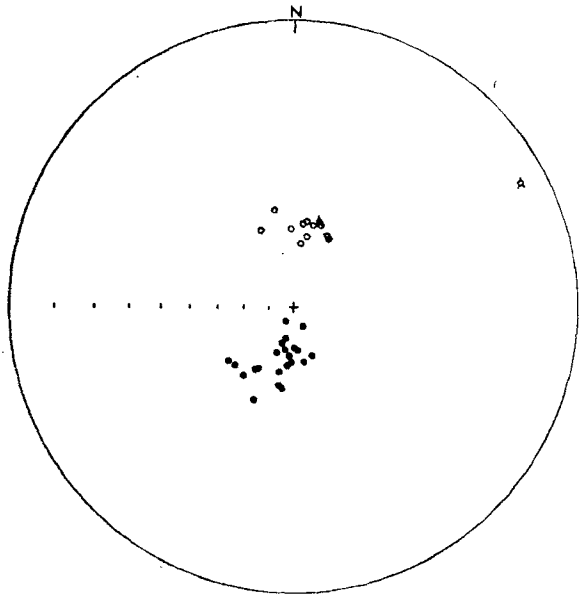



FIG. 8. Barrington volcano. Stereographic projection of directions of magnetization of all stable flows. Symbols are as in Fig. 4.

## 4. Frequency of reversals in the Cenozoic

The time of polarity changes of the geomagnetic field for the past few million years has been found by a combination of K–Ar dating and palaeomagnetic studies on igneous rocks (McDougall & Chamalaun 1966; Cox, Dalrymple & Doell 1967). This time scale is consistent with that determined from the direction of magnetization of ocean cores (Hays & Opdyke 1967). The symmetrical and parallel nature of the magnetic anomalies on the flanks of the mid-ocean ridges (Pitman, Herron & Heirtzler 1968; Dickson, Pitman & Heirtzler 1968; Le Pichon & Heirtzler 1968) conforms to the crustal model predicted by Vine & Matthews (1963). This is regarded as strong evidence for the concept of sea-floor spreading (Hess 1962). Assuming constant spreading rate, the magnetic anomaly pattern associated with the axis of the ridge can always be related to the known sequence of magnetic polarity events for the past 3·35 m.yr. Extrapolating back to about 80 m.yr using data from the South Atlantic, Heirtzler *et al.* (1968) have deduced a geomagnetic polarity time scale for the whole of the Cenozoic. This time scale shows that the frequency of reversals changed about 45 m.yr ago. The recent average rate of three field reversals per m.yr persists back to 45 m.yr, but prior to that a rate of only one per m.yr is deduced. However, the frequency of reversals deduced from this time scale depends largely on the number of ' events ' found in each epoch. In the proposed time scale the reduction in reversal frequency is largely the result of the absence of ' events ' prior to 45 m.yr.

From the observed number of polarity changes in the sections sampled in each volcano and the estimated maximum time covered by each section (Table 4), a minimum estimate can be made of the reversal frequencies at the time of extrusion of the lavas of each volcano. In the Nandewar Volcano (17 m.yr), one reversal is observed in a maximum of 0·6 m.yr or 1·7 polarity changes per m.yr. For the Liverpool Volcano (34 m.yr), three reversals are observed in less than 1·4 m.yr or 2·1 per m.yr and in the Barrington Volcano (52 m.yr), four reversals are recorded in less than 1·4 m.yr or 2·9 per m.yr. As these are minimum estimates, both the

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Nandewar and Liverpool results are perfectly consistent with the predicted time scale, but the frequency observed in the Barrington Volcano is higher than that predicted.

The change from normal to reversed polarity near the top of the Nandewar section (17·5±0·3 m.yr) correlates well with the change (Heirtzler *et al.* 1968) from a normal period (17·80–17·33 m.yr) to a reversed period (17·33–16·41 m.yr) at 17·33 m.yr. The Liverpool section (33·7±0·7 m.yr) consists of a normal 'event' within a reversed period followed by a normal period. This corresponds best with the predicted reversed period from 34·52–34·07 m.yr and a normal period from 34·07–33·61 m.yr, the 'event' not being predicted. The Barrington sections (51·6±0·7 m.yr) contain four polarity changes which fall within the predicted long reversed period from 52·41–49·58 m.yr. This suggests that the detail in the time scale of Heirtzler *et al.* (1968) has been lost prior to 45 m.yr. Although 'events' are not detailed, the time scale may still be a correct record of the epochs for this period.

## 5. Palaeolatitudes and oxygen isotope temperatures

In Table 6 are listed palaeomagnetic poles deduced from studies of Cenozoic and some Mesozoic rocks in eastern Australia. Only those studies utilizing magnetic cleaning techniques are tabulated; these are considered the most reliable available. One result has then been omitted for the Cenozoic, that of Irving *et al.* (1961). Although cleaning techniques were used, the ages of the volcanics studied cover such a wide range (see Introduction) that the results are largely superseded by the new data presented in this paper. Of the 32 sites in the Newer Volcanics of Victoria investigated by Green & Irving (1958), 16 had reversed polarity. Although magnetic cleaning was not carried out, it can be argued that, in the special case where equal numbers of normal and reversed directions are present, the average of these directions taken irrespective of sign is equivalent to magnetic cleaning. However, we have preferred to consider only those results in which magnetic cleaning techniques were used, in spite of the fact that fewer localities (11 sites) are contained in the McDougall *et al.* (1966) survey of the Newer Volcanics. The Tasmanian dolerite survey of Irving (1963) is given as representative of the pre-Cretaceous results.

From these results the palaeolatitude of Canberra for the past 170 m.yr can be deduced on the usual assumption of an axial dipole field. These are shown in Fig. 9(a). The latitude of 72° deduced for Jurassic and Middle Cretaceous times decreased to 39° between 100 m.yr (mid-Cretaceous) and 25 m.yr (Miocene). During the past 25 m.yr the palaeolatitude changes by only 4° to its present value of 35°.

It is interesting to compare the palaeolatitudes deduced above with the past climates of Australia. Sea water temperature (a climatic indicator) is controlled by the total radiated solar energy, the proportion of radiation received (a latitude effect), and minor effects of oceans, continents and mountain ranges. It has a present average gradient of 3·5 °C/° latitude. A latitude change of −37° since the mid-Cretaceous is hence equivalent to about 13 °C increase in sea water temperature, other factors being constant. The sea water temperatures inferred from oxygen isotopes in Australian marine fossils (Dorman & Gill 1959; Dorman 1968) after being reduced to the same latitude, are plotted in Fig. 9(b). Although this shows large temperature variations it does not show the expected trend. It is unlikely that this is because of the ecology and growth characteristics of the animals, since *Ostrea* and *Chlamys* are thought to give temperatures higher than oceanic surface temperatures (Dorman 1968), and belemnites are more likely to give temperatures below oceanic surface temperatures. It seems probable therefore that either the total solar radiation has changed or that other factors are larger than expected.

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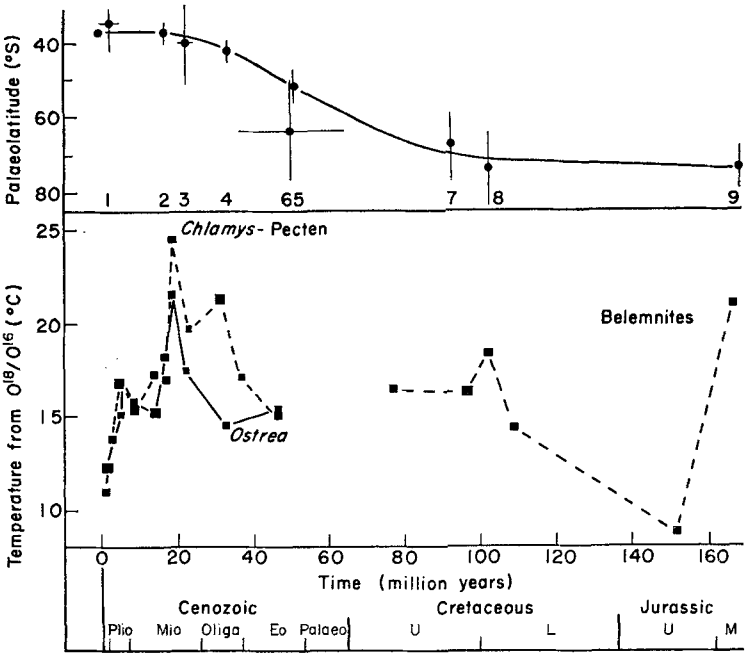



FIG. 9. Palaeolatitude change for Canberra since the mid-Jurassic (top curve). The numbers refer to the data of Table 6 and 95 per cent confidence limits are indicated where known.

Oxygen isotope palaeotemperatures for Australia since the mid-Jurassic (bottom curves) reduced to latitude 38° S. Area of squares is proportional to the number of observations.

## 6. Australian polar-wandering path during the Cenozoic

In Fig. 10 is presented a revised polar-wander path for Australia during the Cenozoic and late Mesozoic. The data points are listed in Table 6, which incorporates the new results presented in this paper. There is an important difference between the new pole path and that derived earlier (Irving *et al.* 1963). The previous results indicated a general progression of the pole from a position just south of Tasmania (in the late Mesozoic) approximately due south along a line between longitudes 120 and 140° E. More recently, results from Miocene igneous rocks in south-east Queensland (Robertson 1966) could be interpreted as suggesting a possible westward excursion of the South Pole during this time. However, this could not be inferred with certainty from the data since the polar error just includes the present South Pole. The more detailed work carried out here both confirms and emphasizes this westward excursion. During the mid-Tertiary there is a pronounced ' knee ' in the revised polar-wander path as shown in Fig. 10. The new results demonstrate that by careful palaeomagnetic measurements and K–Ar dating of sequences of lava flows, the fine structure of Cenozoic polar-wander paths can be found.

The measurements for Pole 6 in Fig. 10 (Older Volcanics of Victoria) are the only ones reported for which K–Ar dates are not yet available. From palaeontological evidence the basalts are regarded as Palaeocene to Eocene (Singleton & Joyce 1969). Its position on the polar-wander path would be most compatible with an Early Palaeocene age, a point which is obvious from the palaeolatitude curve of Fig. 9(a).

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The data suggests that polar movement with respect to Australia for the past 100 m.yr has occurred at a uniform rate (within experimental error) of about 0·45° per m.yr ($\sim$ 5 cm yr$^{-1}$). Recently Irving & Robertson (1969) have suggested that one-half of the polar movement took place over a short period about 25 m.yr ago. The data now refutes this suggestion. However it is still possible that a period of more rapid movement occurred between 100 and 50 m.yr ago. We consider this to be unlikely.

Le Pichon & Heirtzler (1968) have analysed the magnetic anomalies across the mid-ocean ridge between Australia and Antarctica. Their analysis indicates a spreading rate of 3 cm yr$^{-1}$ at the ridge, and that the anomalies terminate at the Australian continental shelf at anomaly 17, estimated to be 43 m.yr ago (Heirtzler *et al.* 1968). Prior to that time Australia and Antarctica were joined together and have subsequently drifted apart. Assuming the ridge to be fixed this indicates that Australia drifted 15° northwards approximately along a present meridian during the past 43 m.yr. If this is the only process which has occurred in this time, the movement of the pole should reflect this precisely. The observed polar movement for the past 43 m.yr, however, does not show the expected change. Instead, there is in addition a predominantly westward component until 25 m.yr ago followed by a predominantly eastward component up to the present. These additional components are presumably due to polar-wandering as distinct from continental drift. These components should be a common feature of the polar-wander curves of all the continents. However, detailed studies of Tertiary rocks from other continents in the manner undertaken in this study have not yet been reported. McElhinny *et al.* (1968) in their

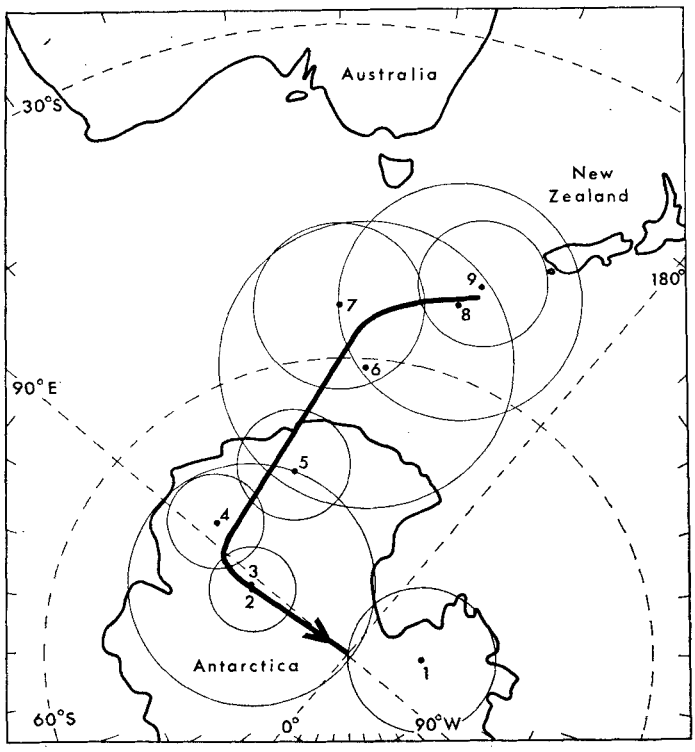



FIG. 10. Polar-wander curve for Australia since the mid-Jurassic. The poles are numbered as in Table 6 and circles of 95 per cent confidence are indicated.

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summary of African palaeomagnetic results suggest a possible similar westward excursion of the south pole during the Cenozoic. However, this suggestion was based upon results of a preliminary nature only and awaits confirmation from detailed work now in progress (Brock 1968).

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## APPENDIX I

*Mean Flow Direction of Magnetization after Magnetic Cleaning.*

| Flow no. | Altitude | *D*(deg.) | *I*(deg.) | Polarity |
|---|---|---|---|---|
| | | Nandewar Volcano | | |
| *Lindesay Creek* | | | | |
| 48 | 1421 | 196·5 | +42·5 | R |
| 47 | 1385 | 197·3 | +62·0 | R |
| 46 | 1347 | 190·6 | +67·3 | R |
| 45 | 1332 | 012·3 | −33·8 | |
| 44 | 1321 | 027·1 | −43·3 | |
| 43 | 1317 | 027·7 | −42·0 | |
| 42 | 1302 | 002·1 | −25·1 | |
| 41 | 1290 | 108·5 | −56·5 | |
| 40 | 1287 | 011·0 | −59·0 | |
| 39 | 1280 | 070·0 | −81·1 | |
| 38 | 1276 | 114·8 | −81·2 | |
| 37 | 1251 | 169·5 | −76·5 | |
| 36 | 1248 | 127·7 | −71·3 | |
| 35 | 1241 | 162·0 | −69·1 | |
| 34 | 1222 | 014·8 | −63·7 | N |
| 33 | 1219 | 010·8 | −62·8 | N |

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## Appendix I—*continued*

| Flow no. | Altitude | *D*(deg.) | *I*(deg.) | Polarity |
|---|---|---|---|---|
| 32 | 1214 | 029·5 | −61·0 | N |
| 31 | 1207 | 022·0 | −51·8 | N |
| 30 | 1202 | 015·1 | −49·0 | N |
| 29 | 1194 | 019·5 | −50·1 | N |
| 28 | 1183 | 004·0 | −32·8 | N |
| 27 | 1174 | 352·7 | −55·5 | N |
| 26 | 1169 | 356·5 | −53·0 | N |
| 25 | 1161 | 344·1 | −50·6 | N |
| 24 | 1157 | 351·8 | −45·6 | N |
| 23 | 1152 | 351·5 | −43·8 | N |
| 22 | 1147 | 349·0 | −52·5 | N |
| 21 | 1143 | 352·0 | −53·1 | N |
| 20 | 1133 | 014·7 | −31·6 | N |
| 19 | 1124 | 024·7 | −56·8 | N |
| 18 | 1118 | 036·1 | −56·8 | N |
| 17 | 1114 | 033·0 | −44·5 | N |
| 16 | 1111 | 023·7 | −53·8 | N |
| 15 | 1107 | 018·5 | −53·7 | N |
| *14* | *1100* | *028·0* | *−57·5* | *N* |
| 13 | 1096 | 013·2 | −49·0 | N |
| 12 | 1072 | 031·1 | −55·7 | N |
| *Mt Kaputar Road* | | | | |
| 11 | 1065 | 001·1 | −17·5 | |
| 10 | 1042 | 344·7 | −49·1 | N |
| 9 | 952 | 027·2 | −50·0 | N |
| 8 | 908 | 012·8 | −65·0 | N |
| 7 | 885 | 014·1 | −52·0 | N |
| 6 | 885 | 015·6 | −64·2 | N |
| 5 | 850 | 256·2 | −48·9 | |
| 4 | 827 | 344·9 | −27·1 | |
| 3 | 785 | 033·9 | −44·1 | N |
| 2 | 738 | 012·4 | −43·4 | N |
| 1 | 698 | 011·5 | −46·5 | N |
| | | Liverpool Volcano | | |
| 48 | 1160 | 021·6 | −56·6 | N |
| 47 | 1150 | 042·6 | −58·3 | N |
| 46 | 1140 | 018·5 | −48·7 | N |
| 45 | 1130 | 058·2 | −60·7 | N |
| 44 | 1120 | 015·6 | −54·8 | N |
| 43 | 1105 | 094·6 | −82·7 | N |
| 42 | 1063 | 354·9 | −53·2 | N |
| 41 | 1045 | 037·5 | −69·4 | N |
| 40 | 1035 | 019·7 | −77·4 | N |
| 39 | 1008 | 021·1 | −70·4 | N |
| 38 | 955 | 021·9 | −54·7 | N |
| 37 | 925 | 027·5 | −42·7 | N |
| 36 | 923 | 008·4 | −53·6 | N |
| 35 | 920 | 017·5 | −51·4 | N |
| 34 | 890 | 012·0 | −55·5 | N |
| 33 | 865 | 011·6 | −55·8 | N |
| 32 | 860 | 354·8 | −15·4 | |
| 31 | 854 | 015·8 | −60·0 | |
| 30 | 851 | 231·9 | +02·1 | |
| 29 | 725 | 207·4 | +50·1 | R |
| 28 | 723 | 220·0 | +51·2 | R |
| 27 | 721 | 196·5 | +57·8 | R |
| 26 | 707 | 209·4 | +51·6 | R |
| 25 | 705 | 200·1 | +60·5 | R |
| 24 | 701 | 226·6 | +67·6 | R |
| 23 | 698 | 205·4 | +56·8 | R |

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**Appendix I**—*continued*

| Flow no. | Altitude | *D*(deg.) | *I*(deg.) | Polarity |
|---|---|---|---|---|
| 22 | 671 | 203·8 | +57·6 | R |
| 21 | 665 | 207·1 | +57·2 | R |
| 20 | 660 | 195·2 | +60·1 | R |
| 19 | 656 | 195·1 | +63·9 | R |
| 18 | 645 | 199·4 | +61·3 | R |
| 17 | 590 | 019·1 | −55·5 | N |
| 16 | 585 | 333·8 | +22·5 | |
| 15 | 568 | 358·1 | +11·1 | |
| 14 | 552 | 120·7 | +08·1 | |
| 13 | 547 | 110·7 | +71·9 | |
| 12 | 545 | 289·9 | +33·0 | |
| 11 | 514 | 223·4 | +55·9 | |
| 10 | 508 | 231·7 | +47·3 | |
| 9 | 504 | 331·2 | +81·7 | |
| 8 | 500 | 144·1 | +68·1 | |
| 7 | 475 | 186·1 | +62·2 | R |
| 6 | 468 | 179·4 | +58·6 | R |
| 5 | 465 | 179·8 | +66·2 | R |
| 4 | 444 | 182·3 | +57·1 | R |
| 3 | 430 | 192·0 | +62·8 | R |
| 2 | 427 | 161·8 | +55·3 | R |
| 1 | 400 | 223·4 | +37·7 | R |
| | | Barrington Volcano | | |
| *Stewarts Brook* | | | | |
| 35 | 1508 | 158·7 | +69·5 | R |
| 34 | 1495 | 192·0 | +73·1 | R |
| 33 | 1453 | 203·2 | +71·0 | R |
| 32 | 1440 | 182·2 | +68·0 | R |
| 31 | 1410 | 061·5 | −05·5 | |
| 30 | 1400 | 195·5 | +77·2 | R |
| 29 | 1388 | 175·0 | +73·1 | R |
| 28 | 1280 | 189·1 | +58·1 | R |
| 27 | 1220 | 026·7 | −59·6 | N |
| 26 | 1203 | 018·1 | −56·1 | N |
| 25 | 1188 | 009·5 | −56·0 | N |
| 24 | 1178 | 006·0 | −64·3 | N |
| 23 | 1170 | 010·1 | −61·3 | N |
| 22 | 1150 | 348·6 | −51·2 | N |
| 21 | 1116 | 013·5 | −56·7 | N |
| 20 | 1103 | 336·7 | −57·2 | N |
| 19 | 1089 | 357·2 | −58·8 | N |
| 18 | 1083 | 006·0 | −57·1 | N |
| 17 | 1060 | 191·6 | +59·2 | R |
| 16 | 1050 | 016·5 | +54·8 | |
| 15 | 1045 | 187·5 | +66·3 | R |
| 14 | 1030 | 155·0 | +82·2 | R |
| *Sempill Creek* | | | | |
| 13 | 1184 | 193·3 | +74·0 | R |
| 12 | 1124 | 185·8 | +70·9 | R |
| 11 | 1099 | 179·9 | +73·9 | R |
| 10 | 1093 | 199·3 | +75·0 | R |
| 9 | 1060 | 169·7 | +68·0 | R |
| 8 | 1045 | 212·3 | +61·0 | R |
| 7 | 1009 | 217·2 | +56·9 | R |
| 6 | 1001 | 231·2 | +56·9 | R |
| 5 | 984 | 024·4 | −58·7 | N |
| 4 | 952 | 203·7 | +51·2 | R |
| 3 | 928 | 226·0 | +57·5 | R |
| 2 | 917 | 210·6 | +62·1 | R |
| 1 | 885 | 211·6 | +83·6 | R |

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## Appendix II

*K–Ar Samples*

Nandewar Volcano

*Lindesay Creek* 30° 16·8′ S, 150° 09·0′ E. Map sheet SH56–9, Manilla 1 : 250 000, Grid Reference (G.R. 310250–310251.

GA2940 Altitude 1347 m. Fine-grained holocrystalline trachyte with fresh feldspars but the rock has numerous closely spaced cracks.

GA2941 Altitude 1317 m. Fine-grained mugearite with fresh feldspars but 10 per cent brown intergranular material.

GA2942 Altitude 1251 m. Fine-grained holocrystalline hawaiite, with fresh feldspar but 5 per cent brown intergranular material.

*Mount Kaputar Road*

GA2943 Altitude 1363 m. 30° 17·3′ S, 150° 08·9′ E. SH56–9, Manilla 1 : 250 000, G.R. 311249. A fine-grained trachyte with fresh feldspar but 20 per cent brown intergranular material.

GA2914 Altitude 925 m. 30° 16·7′ S, 150° 03·9′ E. SH56–9 Manilla 1 : 250 000, G.R. 301251. A very fine-grained alkali olivine basalt with fresh fedlspars but 10 per cent brown intergranular material.

Liverpool Volcano

*Yarraman Creek.* 31° 44·0′ S, 150° 06·9′ E to 31° 43·5′ S, 150° 09·0′ E. SH56–13, Tamworth 1 : 250 000, G.R. 308075–312075.

GA2915 Altitude 1160 m. Fine-grained holocrystaline alkali olivine basalt, with feldspar cloudy around the margins.

GA3466 Altitude 955 m. Fine-grained alkali olivine basalt with feldspars as poorly defined crystals in the groundmass.

GA3467 Altitude 851 m. Fine-grained alkali olivine basalt with a partly altered groundmass.

GA2913 Altitude 665 m. Medium-grained alkali olivine basalt with cloudy margins to the feldspars.

GA2916 Altitude 400 m. Holocrystalline, coarse-grained, alkali olivine basalt, with feldspars cloudy around their margins.

*Spur from Bundulla Trigonometrical Station to Yarraman Creek.* 31° 44·3′ S, 150° 07·5′ E. SH56–13, Tamworth 1 : 250 000, G.R. 309074.

GA2945 Altitude 547 m. Medium-grained hawaiite, with plagioclase cloudy, at the edges, and some groundmass alteration.

GA2944 Altitude 468 m. Holocrystalline, coarse-grained, alkali olivine basalt with little alteration.

Barrington Volcano

*Sempill Creek.* 31° 51·7′ S, 151° 23·7′ E. SH56–13, Tamworth 1 : 250 000, G.R.441060–442059.

GA3465 Altitude 1330 m. Medium-grained alkali olivine basalt with a poorly crystallized groundmass.

GA2903 Altitude 1184 m. Fine-grained holocrystalline alkali olivine basalt, with no apparent alteration.

GA2923 Altitude 1124 m. Fine-grained holocrystalline alkali olivine basalt with feldspar about 20 per cent cloudy.

GA2902 Altitude 1124 m. Same flow as above. Fine-grained holocrystalline alkali olivine basalt with feldspars about 20 per cent cloudy.

GA2901 Altitude 917 m. Medium-grained holocrystalline alkali olivine basalt with fresh feldspars but 10 per cent altered interstitial material.

GA2900 Altitude 885 m. Medium-grained alkali olivine basalt with fresh feldspars and 30 per cent poorly crystallized groundmass.

*Stewarts Brook.* 32° 02′ S, 151° 23′ E, I56 8/369, Wooloomba 1 : 63 360, G.R. 418375–405378.

GA2924 Altitude 1550 m at Barrington Trigonometrical Station. Holocrystalline medium-grained alkali olivine basalt with no apparent alteration.

GA2925 Altitude 1550 m at Barrington Trigonometrical Station. Holocrystalline medium-grained alkali olivine basalt with no apparent alteration.

GA2949 Altitude 1280 m. Fine-grained holocrystalline alkali olivine basalt with no apparent alteration.

GA2929 Altitude 1116 m. Nepheline basanite composed of 50 per cent 5 mm pyroxene agregates in a coarse-grained groundmass of poorly defined crystals.

GA2948 Altitude 880 m. Outcrop on Stewarts Brook–Mount Barrington fire trail. 32° 00·6′ S, 151° 21·8′ E, I56 8/369, Wooloomba 1 : 63 360, G.R. 375412. Fine-grained holocrystalline alkali olivine basalt with no apparent alteration.

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